



Class:

Mathematics Extension 1

- Attempt Questions 1–7
- All questions are of equal value

[illegible]

Question 1: (12 Marks)**Start a new Answer Sheet**

(a) Differentiate $y = e^x \sin^{-1} x$ **2**

(b) Using the substitution $u = 4 - x^2$, evaluate the following: **3**

$$\int_0^2 \frac{x dx}{\sqrt{4 - x^2}}$$

(c) For the function $y = 2\cos^{-1} 4x$ **2**

i. State the domain and range.

ii. Draw a neat sketch of $y = 2\cos^{-1} 4x$ **2**

(d) Simon is preparing to study for his trial examinations. He has borrowed 3 Mathematics, 5 English and 2 Science books from the library. Each of the books is different.

i. How many ways can he arrange the books on his shelf if there are no restrictions? **1**

ii. How many ways are there of arranging the books on the shelf if a Science book is at each end and the Maths and English books are kept in subject areas. **2**

Question 2: (12 Marks)**Start a new Answer Sheet**

- (a) The acute angle between the tangents to the curves $y = e^{2x}$ and $y = e^{-mx}$ at $x = 0$ is 45° .

Find the values of m .

4

- (b) The equation $2x^3 + 6x^2 + 9x - 2 = 0$ has roots α , β and γ . Find:

i. $\alpha + \beta + \gamma$

1

ii. $\alpha\beta + \beta\gamma + \gamma\alpha$

1

iii. $\alpha^2 + \beta^2 + \gamma^2$

2

- (c) Find the exact value of: $\cos 75^\circ$

2

- (d) Given that $\tan A$ and $\tan B$ are the roots of the equation $2x^2 - 3x - 1 = 0$;

2

Find the value of $\tan(A + B)$

Question 3: (12 Marks)**Start a new Answer Sheet**

(a) Given the function $y = \sin^{-1} \frac{x}{2}$; find:

i. $\frac{dy}{dx}$.

1

ii. The gradient at $x = \sqrt{3}$.

2

iii. The equation of the normal in exact form.

3

(b) i. Differentiate $y = -4x + 4x \log_e 4x$

2

ii. Using the result in (i) find the minimum value for $y = 4x \log_e 4x - 4x$

2

(c) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{2x} \right)$

2

Question 4: (12 Marks)**Start a new Answer Sheet**

- (a) Two points on the parabola $x^2 = 4ay$ are P and Q and have coordinates $(2ap, ap^2)$ and $(2aq, aq^2)$ respectively.
- i. Find the equation of the chord PQ . 2
 - ii. Find the coordinates of the midpoint $M(x, y)$ of PQ . 1
 - iii. If $\angle POQ = 90^\circ$, where O is the origin, prove that $pq = -4$. 1
 - iv. Hence find the locus of M as P and Q move along the parabola. 2
- (b) Use Newton's method to find a second approximation to the positive root of $\sin x + x - 2 = 0$. Take $x = 1.1$ as the first approximation. 2
- (c) Prove by mathematical induction that: 4

$$\sum_{r=1}^n (3r-1) = \frac{3n^2 + n}{2} \quad n \geq 1$$

Question 5: (12 Marks)**Start a new Answer Sheet**

(a) A cross section of a termites mound is found to have equation $y = \frac{15}{8+2x^2}$.

i. Draw a neat sketch of $y = \frac{15}{8+2x^2}$ between $x = -2\sqrt{3}$ and $x = 2\sqrt{3}$. **2**

ii. Calculate the area of the cross-section between $x = -2\sqrt{3}$ and $x = 2\sqrt{3}$. **3**

(b) Solve the equation $\sqrt{3}\sin A + \cos A = 1$ in the domain $0 \leq A \leq 2\pi$ **4**

(c) Using the substitution $x = 3\cos\theta$ show that:

$$\int \frac{x^2 dx}{\sqrt{9-x^2}} = -\frac{9}{2}\cos^{-1}\frac{x}{3} - \frac{1}{2}x\sqrt{9-x^2} + c$$

3

Question 6: (12 Marks)**Start a new Answer Sheet**

(a) For the function $f(x) = \frac{x^2}{x^2-4}$

i. Show the function is an even function.

1

ii. State any vertical asymptotes.

1

iii. Find the horizontal asymptotes.

1

iv. Find where the graph cuts the y axis.

1

v. Draw a neat sketch of the function $f(x) = \frac{x^2}{x^2-4}$

2

(b) Evaluate $\int_0^{\frac{\pi}{3}} \sin^2 x \, dx$

2

(c) The perpendicular height of a jet above the ground is 2000 metres. An observer due east of the jet looks up at the jet at an angle of elevation of 60° . A second person looks up at the jet at an angle of elevation of 30° . If the two people subtend an angle of 80° at the base of the perpendicular below the jet, calculate:

i. The exact length of BD and BC.

2

ii. The distance between the two people, to the nearest metre.

2

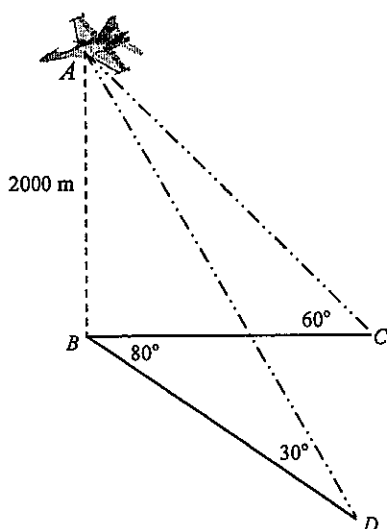
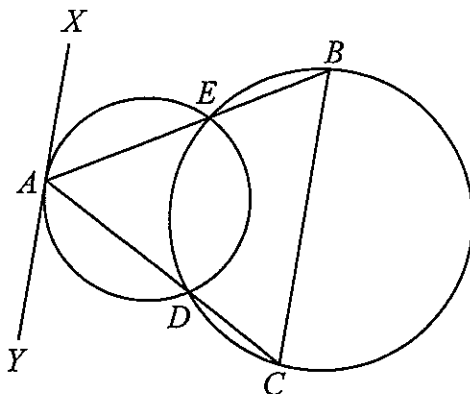


Diagram not to scale.

Question 7: (12 Marks)**Start a new Answer Sheet**

- (a) Two circles intersect at E and D . From a point A two lines are drawn through the points of intersection to meet the other circle at B and C as shown in the diagram.

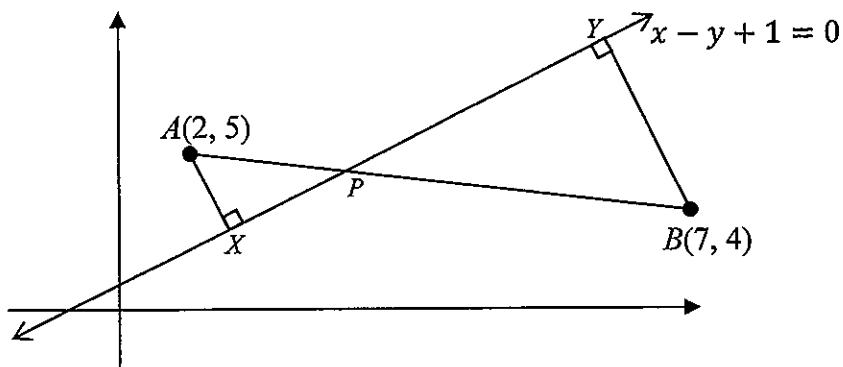
Prove the tangent XY at A is parallel to the line BC .

3

- (b) Solve the inequation $\left|2x - \frac{1}{2}\right| > \sqrt{x - x^2}$

4

- (c) The points $A(2, 5)$ and $B(7, 4)$ are on either side of the line; $x - y + 1 = 0$.



- i. Prove $\triangle AXP \parallel \triangle BYP$
- ii. **Hence** find the coordinates of P that divides the interval AB .

2**3****End of Examination**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Solutions Trial HSC Mathematics Extension 1

Question 1

$$(a) \frac{d}{dx}(e^x \sin^{-1} x) = (\sin^{-1} x)(e^x) + (e^x) \left(\frac{1}{\sqrt{1-x^2}} \right)$$

$$\frac{d}{dx}(e^x \sin^{-1} x) = e^x \left[(\sin^{-1} x) + \left(\frac{1}{\sqrt{1-x^2}} \right) \right]$$

2 Marks – correct answer

1 Mark – correctly finding u' and v'

$$(b) u = 4 - x^2$$

$$\frac{du}{dx} = -2x$$

$$-\frac{du}{2} = x dx$$

1 Mark

$$x = 2 \quad u = 0$$

$$x = 0 \quad u = 4$$

$$\int_0^2 \frac{x dx}{\sqrt{4-x^2}} = -\frac{1}{2} \int_4^0 \frac{du}{\sqrt{u}}$$

$$\int_0^2 \frac{x dx}{\sqrt{4-x^2}} = \frac{1}{2} \int_0^4 u^{-\frac{1}{2}} du$$

$$\frac{1}{2} \int_0^4 u^{-\frac{1}{2}} du = \frac{1}{2} [2\sqrt{u}]_0^4$$

1 Mark

$$\frac{1}{2} [2\sqrt{u}]_0^4 = \frac{1}{2} [(4) - (0)]$$

$$\therefore \int_0^2 \frac{x dx}{\sqrt{4-x^2}} = 2$$

1 Mark

$$(c) y = 2\cos^{-1} 4x$$

$$i. \quad D: -1 \leq 4x \leq 1$$

1 Mark

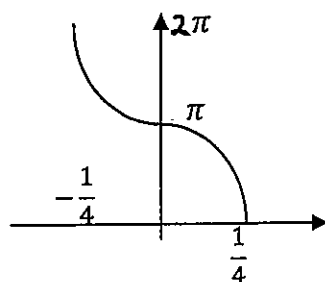
$$D: -\frac{1}{4} \leq x \leq \frac{1}{4}$$

$$R: 2(0) \leq y \leq 2(\pi)$$

1 Mark

$$R: 0 \leq y \leq 2\pi$$

1 Mark – correct graph



1 Mark – labelling axes

(d) i. No restrictions = 10!

1 Mark

ii. $N^0 \text{ ways} = (2!)[(2!)(3!)(5!)]$?

2 Marks – correct working and answer

1 Mark – some correct working with explanation

Question 2

(a)

$$y = e^{2x}$$

$$\frac{dy}{dx} = 2e^{2x} \text{ at } x = 0$$

$$m_1 = 2$$

$$y = e^{-mx}$$

$$\frac{dy}{dx} = -me^{-mx} \text{ at } x = 0$$

1 Mark

$$m_2 = -m$$

$$\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45 = \left| \frac{2 + m}{1 - 2m} \right|$$

$$1 = \left| \frac{2 + m}{1 - 2m} \right|$$

1 Mark

$$\frac{2 + m}{1 - 2m} = 1$$

$$\frac{2 + m}{1 - 2m} = -1$$

$$2 + m = 1 - 2m$$

$$2 + m = 2m - 1$$

1 Mark

$$3m = -1$$

$$3 = m$$

$$m = -\frac{1}{3}$$

$$m = 3$$

1 Mark

Since the gradient must be negative $m = -\frac{1}{3}$

(b) $2x^3 + 6x^2 + 9x - 2 = 0$

i. $\alpha + \beta + \gamma = \frac{-6}{2} = -3$

1 Mark

ii. $\alpha\beta + \beta\gamma + \alpha\gamma = \frac{9}{2}$

1 Mark

iii. $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$

$$= (-3)^2 - 2\left(\frac{9}{2}\right)$$

1 Mark

$$= 0$$

1 Mark

$$(c) \cos(30 + 45) = \cos 30 \cos 45 - \sin 30 \sin 45$$

1 Mark

$$\cos(30 + 45) = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) - \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right)$$

$$\cos(30 + 45) = \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) - \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

$$\cos(30 + 45) = \frac{\sqrt{6} - \sqrt{2}}{4}$$

1 Mark

$$(d) 2x^2 - 3x - 1$$

$$\alpha = \tan A \text{ and } \beta = \tan B$$

$$\alpha + \beta = \tan A + \tan B = \frac{3}{2}$$

$$\alpha\beta = \tan A \tan B = -\frac{1}{2}$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A + B) = \frac{\frac{3}{2}}{1 - \left(-\frac{1}{2}\right)}$$

1 Mark

$$\tan(A + B) = \frac{\left(\frac{3}{2}\right)}{\left(\frac{3}{2}\right)}$$

$$\tan(A + B) = 1$$

1 Mark

Question 3

(a)

$$i. \quad \frac{dy}{dx} = \frac{1}{\sqrt{4-x^2}}$$

1 Mark

$$ii. \quad m = \frac{1}{\sqrt{4-(\sqrt{3})^2}}$$

1 Mark

$$m = 1 \text{ (tangent)}$$

1 Mark

$$m = -1 \text{ (normal)}$$

$$iii. \quad \text{At } x = \sqrt{3}, y = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3}$$

1 Mark

$$y - \frac{\pi}{3} = -1(x - \sqrt{3})$$

1 Mark

$$3y - \pi = -3x + 3\sqrt{3}$$

$$3x + 3y - 3\sqrt{3} - \pi = 0$$

1 Mark

$$(b) \frac{dy}{dx} = \left[(\log_e 4x)(4) + (4x) \left(\frac{4}{4x} \right) \right] - 4$$

1 Mark

$$\frac{dy}{dx} = 4 \log_e 4x$$

1 Mark

(c) Min value will occur at the turning point

$$\frac{dy}{dx} = 4 \log_e 4x$$

$$4 \log_e 4x = 0$$

$$\log_e 4x = 0$$

$$e^{\log_e 4x} = e^0$$

$$4x = 1$$

$$x = \frac{1}{4}$$

1 Mark

$$\text{When } x = \frac{1}{4}, y = 4 \left(\frac{1}{4} \right) \log_e \left(4 \cdot \frac{1}{4} \right) - 4 \left(\frac{1}{4} \right) = -1$$

Therefore the minimum value is $y = -1$

1 Mark

(d)

$$\lim_{x \rightarrow 0} \left(\frac{\sin 3x}{2x} \right) = \left(\frac{3}{2} \right) \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \right)$$

1 Mark

$$= \frac{3}{2} (1)$$

$$= \frac{3}{2}$$

1 Mark

Question 4

(a)

$$\begin{aligned} \text{i. } \frac{y - ap^2}{x - 2ap} &= \frac{aq^2 - ap^2}{2aq - 2ap} \\ \frac{y - ap^2}{x - 2ap} &= \frac{a(q-p)(q+p)}{2a(q-p)} \end{aligned}$$

1 Mark

$$y - ap^2 = \frac{(q+p)}{2} [x - 2ap]$$

$$y - ap^2 = \frac{(q+p)x}{2} - 2apq - ap^2$$

$$y = \frac{(q+p)x}{2} - \cancel{2apq} - \cancel{ap^2}$$

1 Mark

$$\text{ii. Midpoint} = \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$\text{Midpoint} = \left(a(p + q), \frac{ap^2 + aq^2}{2} \right)$$

1 Mark

$$\text{iii. } m_1 = \frac{ap^2-0}{2ap-0} = \frac{p}{2} \text{ and likewise } m_2 = \frac{q}{2}$$

$$m_1 \times m_2 = -1$$

$$\frac{p}{2} \times \frac{q}{2} = -1$$

$$pq = -4$$

1 Mark

$$\text{iv. } x = a(p + q)$$

$$\frac{x}{a} = (p + q)$$

$$\left(\frac{x}{a}\right)^2 = (p + q)^2 \dots (1)$$

$$y = \frac{a}{2}(p^2 + q^2) \dots (2)$$

$$y = \frac{a}{2}[(p + q)^2 - 2pq]$$

$$y = \frac{a}{2}\left[\left(\frac{x}{a}\right)^2 - 2(-4)\right]$$

$$y = \frac{x^2}{2a} + 4a$$

$$2ay = x^2 + 8a^2$$

$$2ay - 8a^2 = x^2$$

Therefore the locus is:

$$x^2 = 2a(y - 4a)$$

**1 Mark – correctly
sub (1) into (2)**

1 Mark - answer

(b)

$$x_2 = (1.1) - \frac{\sin(1.1) + 1.1 - 2}{1 \cos(1.1) + 1}$$

$$x_2 = (1.1) - \frac{(-0.00879 \dots)}{-(0.5464 \dots)} \quad 1.4536$$

$$x_2 = (1.1) + 0.01608 \dots$$

$$x_2 = 1.11608 \dots \quad 1.106$$

$$x_2 = 1.11$$

1 Mark

1 Mark

$$(c) \ 2 + 5 + \dots + 3n - 1 = \frac{3n^2 + n}{2}$$

We need to prove that $S_{n+1} = S_n + T_{n+1}$

Step 1: Show true for $n = 1$

$$LHS = 2 \qquad RHS = \frac{3+1}{2} = 2$$

1 Mark

Step 2: Assume true for $n = k$

$$(a) \ 2 + 5 + \dots + 3k - 1 = \frac{3k^2 + k}{2}$$

1 Mark

Step 2: Prove true for $n = k + 1$

$$S_{k+1} = \frac{3(k+1)^2 + (k+1)}{2}$$

$$S_n + T_{n+1} = \frac{3k^2 + k}{2} + 3(k+1) - 1$$

$$S_n + T_{n+1} = \frac{3k^2 + k + 6k + 6 - 2}{2}$$

$$S_n + T_{n+1} = \frac{3k^2 + 7k + 4}{2}$$

$$S_n + T_{n+1} = \frac{3k^2 + 6k + 3 + k + 1}{2}$$

$$S_n + T_{n+1} = \frac{(3k^2 + 6k + 3) + (k+1)}{2}$$

$$S_n + T_{n+1} = \frac{(3(k+1)^2) + (k+1)}{2}$$

$$\therefore S_{n+1} = S_n + T_{n+1}$$

2marks

~~1 Mark~~

Step 4: Conclusion

Hence if the statement is true for $n = k$, then it is also true when $n = k + 1$. The statement is true for $n = 1$ and so it is true for $n = 2$ and so on. Hence it is true for all n .

~~1 Mark~~

Question 5:

- i. The function is even as $f(-x) = f(x)$

$$\lim_{x \rightarrow \infty} \left(\frac{\frac{15}{x^2}}{\frac{8}{x^2} + 2} \right) = \frac{0}{2} = 0$$

$$x = 0, y = \frac{15}{8}$$

- ii.

$$A = 2 \int_0^{2\sqrt{3}} \frac{15}{2(4+x^2)} dx$$

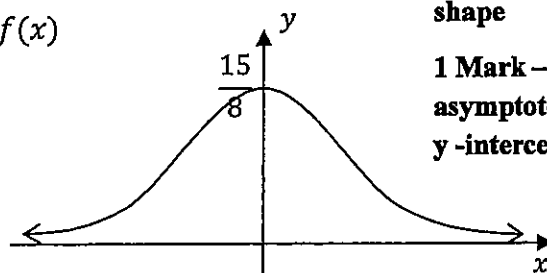
$$A = 15 \int_0^{2\sqrt{3}} \frac{1}{4+x^2} dx$$

$$A = 15 \left[\frac{1}{2} \tan^{-1} \left(\frac{x}{2} \right) \right]_0^{2\sqrt{3}}$$

$$A = 15 \left[\left(\frac{1}{2} \right) \tan^{-1}(\sqrt{3}) - \left(\frac{1}{2} \right) \tan^{-1}(0) \right]$$

$$A = 15 \left[\left(\frac{1}{2} \right) \left(\frac{\pi}{3} \right) - \left(\frac{1}{2} \right) (0) \right]$$

$$\therefore A = \frac{15\pi}{6} \text{ units}^2 \quad \text{or} \quad \frac{5\pi}{2} \text{ or } 7.85$$



1 Mark – correct shape

1 Mark – showing asymptote and y-intercept

1 Mark

1 Mark

1 Mark

$$(b) R = \sqrt{(\sqrt{3})^2 + 1^2} = 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

$$\sqrt{3} \sin A + \cos A \equiv 2 \sin \left(A + \frac{\pi}{6} \right) = 1$$

$$\sin \left(A + \frac{\pi}{6} \right) = \frac{1}{2}$$

1st Quad.

$$\left(A + \frac{\pi}{6} \right) = \sin^{-1} \frac{1}{2}$$

$$\left(A + \frac{\pi}{6} \right) = \frac{\pi}{6}$$

$$A_1 - \frac{\pi}{6} = \frac{\pi}{6}$$

$$\left(A_2 + \frac{\pi}{6} \right) = \pi - \frac{\pi}{6}$$

$$A_1 = 0$$

$$A_2 = \frac{2\pi}{3}$$

Check $x = 2\pi$

$$\sqrt{3} \sin 2\pi + \cos \pi = 0 + 1 = 1 \quad \checkmark$$

1 Mark - A_1 and A_2

Therefore the solutions are: $A = 0, \frac{2\pi}{3}$ and 2π

1 Mark = 2π

(c)

$$x = 3\cos\theta$$

$$\frac{dx}{d\theta} = -3\sin\theta$$

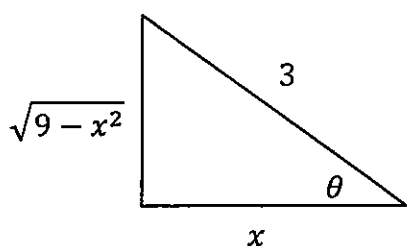
$$dx = -3\sin\theta d\theta$$

1 Mark – finding needed expressions

$$x^2 = 9\cos^2\theta$$

$$x = 3\cos\theta$$

$$\theta = \cos^{-1}\left(\frac{x}{3}\right)$$



$$\int \frac{x^2}{\sqrt{9-x^2}} dx = \int \frac{9\cos^2\theta}{\sqrt{9-9\cos^2\theta}} (-2\sin\theta) d\theta$$

$$= \int \frac{9\cos^2\theta}{\sqrt{9\sin^2\theta}} (-3\sin\theta) d\theta$$

$$= \int \frac{9\cos^2\theta}{3\sin\theta} (-3\sin\theta) d\theta$$

$$= \int -9\cos^2\theta d\theta$$

$$= -9 \left[\frac{\theta}{2} + \frac{1}{4} \sin 2\theta \right] + c$$

1 Mark – correct integration

$$= -9 \left[\frac{\theta}{2} + \left(\frac{1}{4}\right) (2) \sin\theta \cos\theta \right] + c$$

$$= -9 \left[\frac{1}{2} \cos^{-1}\left(\frac{x}{3}\right) + \frac{1}{2} \left(\frac{\sqrt{9-x^2}}{3}\right) \left(\frac{x}{3}\right) \right] + c$$

$$= -\frac{9}{2} \cos^{-1}\left(\frac{x}{3}\right) - \frac{x}{2} \left(\frac{\sqrt{9-x^2}}{3}\right) + c$$

1 Mark – correct answer

Question 6

(a)

i. $f(-x) = \frac{(-x)^2}{(-x)^2 - 4}$

$$f(-x) = \frac{x^2}{x^2 - 4}$$

$$f(-x) = f(x)$$

1 Mark

ii. $x \neq \pm 2$

1 Mark

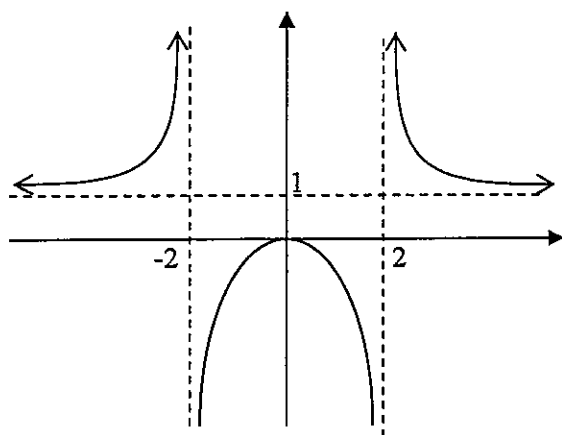
iii. $\lim_{x \rightarrow \infty} \left(\frac{\frac{x^2}{x^2}}{\frac{x^2}{x^2} - 4} \right) = \frac{1}{1-0} = 1$

1 Mark

iv. $x = 0, y = 0$

1 Mark

v.



1 Mark – correct graph

1 Mark – labelling

(c)

i. $\tan 60 = \frac{2000}{BC}$
 $BC = \frac{2000}{\tan 60}$
 $BC = \frac{2000}{\sqrt{3}}$

$\tan 30 = \frac{2000}{BD}$
 $BD = 2000\sqrt{3}$

1 Mark - BC

1 Mark - BD

ii. $(DC)^2 = \left(\frac{2000}{\sqrt{3}}\right)^2 + (2000\sqrt{3})^2 - 2\left(\frac{2000}{\sqrt{3}}\right)(2000\sqrt{3})\cos 80$

1 Mark - BC

iii. $(DC)^2 = 11944.1479 \dots$
 $DC = 3456 \text{ m}$

1 Mark - BC

(b)

$$\int_0^{\frac{\pi}{3}} \sin^2 x dx = \left[\frac{x}{2} - \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{3}}$$

$$\left[\frac{x}{2} - \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{3}} = \left[\left(\frac{\pi}{6} - \frac{1}{4} \left(\frac{\sqrt{3}}{2} \right) \right) \right]$$

$$\int_0^{\frac{\pi}{3}} \sin^2 x dx = \left(\frac{4\pi - 3\sqrt{3}}{24} \right)$$

Question 7

(a)

$$\angle YAD = \angle AED = x \text{ (angle b/w a tangent and chord equals the angle in the alt. segment)}$$

1 M

$$\angle BED = 180 - x \text{ (straight line = 180)}$$

$$\angle BCD = x \text{ (EBCD is a cyclic quad, opposite angle of cyclic quad are supplementary.)}$$

1 M

$$\therefore \angle YAD = \angle BCD = x$$

$$\therefore XY \parallel BC \text{ (Alternate angles)}$$

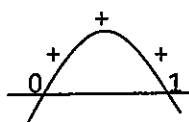
1 Mark

(b) Domain of $\sqrt{x - x^2}$

$$x - x^2 \geq 0$$

$$x(1 - x) \geq 0$$

$$D: 0 \leq x \leq 1$$



1 Mark

$$\left| 2x - \frac{1}{2} \right| = \sqrt{x - x^2}$$

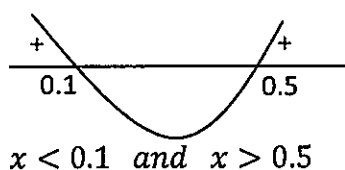
$$\left(2x - \frac{1}{2} \right)^2 = (\sqrt{x - x^2})^2$$

$$4x^2 - 2x + \frac{1}{4} = x - x^2$$

1 Mark

$$20x^2 - 12x + 1 > 0$$

$$(10x - 1)(2x - 1) > 0$$



1 Mark

$$x < 0.1 \text{ and } x > 0.5$$

Putting both answers together we get:

$$0 \leq x < \frac{1}{10} \text{ and } \frac{1}{2} < x \leq 1$$

1 Mark

(c)

$$\angle AXP = \angle BYP = 90^\circ \text{ (given)}$$

1 Mark

$$\angle APX = \angle BPY \text{ (Vert. opp angles are equal)}$$

1 Mark

$$\therefore \triangle XAP \parallel \triangle BYP$$

$$d_{p(AX)} = \frac{|2-5+1|}{\sqrt{2}}$$

$$d_{p(AX)} = \frac{2}{\sqrt{2}} = \sqrt{2} \text{ likewise } d_{p(YB)} = 2\sqrt{2}$$

1 Mark

Therefore the ratio of $AP:PB$ is 1:2 (sides of similar triangles are in same ratio)

1 Mark

$$P\left(\frac{1(7)+2(2)}{3}, \frac{1(4)+2(5)}{3}\right)$$

1 Mark

$$\therefore P\left(\frac{11}{3}, \frac{14}{3}\right)$$

Q3

$$a) (i) \frac{1}{\sqrt{1 - \frac{x^2}{4}}} \times \frac{1}{2}$$

$$y' = \frac{1}{\sqrt{4 - x^2}}$$

$$(ii) y' = 1 \quad 2$$

$$(iii) m_2 = -1$$

$$y - \frac{\pi}{3} = -(x - \sqrt{3}) \quad 3$$

$$y = -x + \sqrt{3} + \frac{\pi}{3} \quad \checkmark$$

$$3x + 3y - 3\sqrt{3} - \pi$$

b)

$$u = 4x = u' = 4$$

$$v = \ln 4x \quad v' = \frac{1}{x}$$

$$y' = 4 \ln 4x + 4 - 4$$

$$y' = 4 \ln 4x = 0$$

$$(iv) x = \frac{1}{4} \quad y = -1 \quad \checkmark$$

$$y'' = \frac{4}{x}$$

2

$$y''(\frac{1}{4}) = 16 > 0 \quad \therefore \text{Min Val at } (\frac{1}{4}, -1)$$

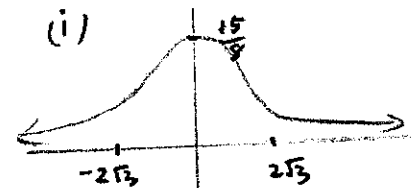
(c)

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{2}$$

$$= \frac{3}{2} \quad 2$$

Q5

$$a) y = \frac{15}{8 + 2x^2}$$



$$A = 2 \int_0^{2\sqrt{3}} \frac{15}{2} \cdot \frac{dx}{4 + x^2}$$

$$= 15 \int_0^{2\sqrt{3}} \frac{1}{2} \tan^{-1} \frac{x}{2} \Big|_0^{2\sqrt{3}}$$

$$= \frac{15}{2} \left(\frac{\pi}{3} \right) = \frac{5\pi}{2} \quad 3$$

$$\div 7.85$$

(b)

$$R = \sqrt{4} = 2$$

$$\tan \alpha = \frac{1}{\sqrt{3}} = \frac{\pi}{6}$$

$$2 \sin(A + \frac{\pi}{6}) = 1$$

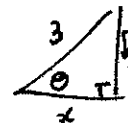
$$A + \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6} \quad 4$$

$$A = 0, \frac{4\pi}{6}, 2\pi$$

$$3\frac{1}{2}$$

$$\cos^{-1} \frac{x}{3} = \theta$$

$$\frac{x}{3} = \cos \theta$$



$$(c) \frac{dx}{d\theta} = -3 \sin \theta$$

$$= \int \frac{9 \cos^2 \theta \cdot -3 \sin \theta d\theta}{\sqrt{9 - 9 \cos^2 \theta}}$$

$$= \int \frac{-27 \cos^2 \theta \sin \theta \cdot d\theta}{3 \sin \theta}$$

$$= -9 \int \cos^2 \theta d\theta$$

$$= \frac{-9}{2} \int 1 + \cos 2\theta d\theta$$

$$= \frac{-9}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) + C$$

$$= \frac{-9}{2} \left(\cos^{-1} \frac{x}{3} + \frac{x}{3} \cdot \frac{\sqrt{9-x^2}}{3} \right) + C$$

$$3$$

$$= \frac{-9}{2} \cos^{-1} \frac{x}{3} - \frac{1}{2} x \sqrt{9-x^2} + C$$